MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 300 VOLTS, TYPE DLT

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

## Construction (non-watertight)

### Insulated steel core member

First - One steel strength member in accordance with MIL-W-83420, 7 by 19 stranding, 0.156-inch finished diameter (not preformed).
Second - 60-percent natural rubber insulation (ASTM D 353). Minimum thickness 0.046 inch, to a nominal diameter of 0.262 inch. The insulation shall fill all outer interstices of the steel strands; this may be applied in one or two layers.

### Insulated conductors (four)

First - Copper conductor. AWG size 18, class M stranding, coated.
Second - Separator.
Third - 30-percent natural rubber insulation, (ASTM D 353). Nominal thickness 0.031 inch. Standard identification code applied by method 3.

### Assembly

First - The four insulated conductors shall be cabled in a close helix about the steel strength member, with a lay of approximately 0.625 inch, partially embedded in the relatively soft partially vulcanized insulation of the steel core member.
Second - Synthetic rubber filler, integral with an 0.031-inch nominal thickness for an insulating belt over the assembly.
Third - Reinforcement under cable jacket.

FSC 6145

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited



Source: https://assist.dla.mil -- Downloaded: 20261O-16T12:05Z
Check the source to verify that this is the current version before use.

Fourth - Polychloroprene jacket (black) minimum thickness 0.078 inch, bonded to the insulating belt. The assembled cable shall be vulcanized into a compact unit. The insulating belt shall be removable from the conductors and the conductors shall be separable without damage to the insulation and without destroying the identity of the color code. Manufacturer's identification shall include molded markings on the jacket at intervals not greater than 25 feet, showing cable type, manufacturer's name or identification, and year of manufacture. Maximum overall diameter 0.710 inch.

EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor resistance - ohms/1000 feet, at 25°C, maximum | 7.15 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2000 |
| Conductor to water (strength member and water bath) | 2000 |
| Insulation resistance - megohms/1000 feet, minimum | |
| Conductor to conductor | 200 |
| Conductor to water (strength member and water bath) | 100 |
| Group A: | |
| Visual and dimensional | No failure |
| Group B: | |
| Physicals (unaged) | |
| Insulation (conductor) (ASTM D 353) | |
| Tensile strength - $lbf/in^2$, minimum | 900 |
| Elongation - percent, minimum | 300 |
| Jacket (cable) | |
| Tensile strength - $lbf/in^2$, minimum | 1800 |
| Elongation - percent, minimum | 300 |
| Bending endurance - at 25°C, on 2-inch maximum diameter mandrel, cycles, minimum | 1000 |
| Group C: | |
| Flammability - inches, maximum | 2 |
| Steel strength member | MIL-W-83420 |
| Physicals (aged) | |
| Insulation (conductor) (ASTM D 353) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 75 |
| Elongation - percent of unaged, minimum | 75 |

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:05Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

| | Requirements |
|---|---|
| Oxygen pressure | |
| Tensile strength - percent of unaged, minimum | 65 |
| Elongation - percent of unaged, minimum | 65 |
| Jacket (cable) | |
| Air pressure heat | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

| | |
|---|---|
| Stress endurance | No failure |

UNIT ORDERING LENGTHS:

The DLT cable shall be delivered in nominal lengths of not less than 600 feet and not more than 630 feet. Except where authorized by the contract or order, not more than 30 percent of the total footage of each lot offered for inspection may be accepted in random lengths of not less than 400 feet, and not more than 420 feet or in random lengths of not less than 200 feet and not more than 210 feet. Combinations thereof of random lengths will be permitted. Other lengths are not suitable for the purpose.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Army - CR, ER
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0885-3)

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:05Z
Check the source to verify that this is the current version before use.